

Algorithms

Chapter 1 Preliminaries

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Outline

- ▶ **Mathematical Notions and Terminology** 數學符號與術語
- ▶ Definitions, Theorems, and Proofs
- ▶ Types of Proof 建構法, 矛盾法, 歸納法

Sets_{1/3}

- ▶ A **set** is a group of objects represented as a unit.
將一群相同性質的事或物用一個群組表示，稱此群組為集合
- ▶ Sets may contain any type of object, including numbers, symbols, and even other sets. 集合中可以有集合
- ▶ The objects in a set are called its **elements** or **members**.
- ▶ One way to describe sets formally is by listing its elements inside braces. 用大括號將集合中的元素括起
(表示集合的方式 ⇒ 列舉)
- ▶ Thus the set $\{7, 21, 57\}$ contains the elements 7, 21, and 57.

Sets_{2/3}

屬於 不屬於

- ▶ The symbols $\in$ and $\notin$ denote set membership and non-membership, respectively.
- ▶ We write $7 \in \{7, 21, 57\}$ and $8 \notin \{7, 21, 57\}$.
- ▶ For two sets A and B , we say that A is a **subset** of B , written $A \subseteq B$, if every member of A is also a member of B .
A 是 B 的子集 $\Rightarrow$ 每一個在 A 的 element 也一定在 B 裡 ($A \subseteq B$)
- ▶ We say that A is a **proper subset** of B , written $A \subsetneq B$, if A is a subset of B and not equal to B . $\hookrightarrow$ 真子集
A 是 B 的子集, 但 $A \neq B$
- ▶ Let $A = \{7, 21\}$ and $B = \{7, 21, 57\}$. Then, we can write $A \subseteq B$ and $A \subsetneq B$.
- ▶ The set of **natural numbers** N is $\{1, 2, 3, \dots\}$. 自然數的集合

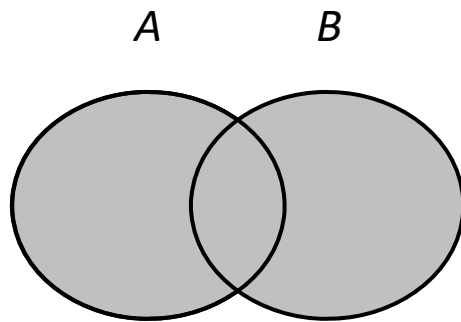
Sets_{3/3}

- ▶ The set with 0 members is called the **empty set** and is written $\emptyset$.
集合中沒有任何元素即為空集合 ($\{\} = \emptyset$) 且空集合為任何集合的子集
 - ▶ A set containing elements according some rule is denoted by $\{n \mid \text{rule about } n\}$. (表示集合的方法 II) $\{n \mid \text{關於 } n \text{ 的規則}\}$
 - ▶ $\{n \mid n = m^2 \text{ for some } m \in N\} \rightarrow$ the set of perfect squares. 完全平方數
Ex: $\{c \mid c \in N, 1 \leq c \leq 5\} = \{1, 2, 3, 4, 5\}$
 - ▶ The **union** of A and B , written $A \cup B$, is the set we get by combining all the elements in A and B into a single set.
聯集: 將 A 和 B 合併形成一個新集合
 - ▶ The **intersection** of A and B , written $A \cap B$, is the set of elements that are in both A and B .
交集: 取共同的元素
 - ▶ The **complement** of A , written $\bar{A}$, is the set of all elements under consideration that are **not** in A .
補集: 所有不在 A 之中的元素所形成的集合
-

Venn diagram

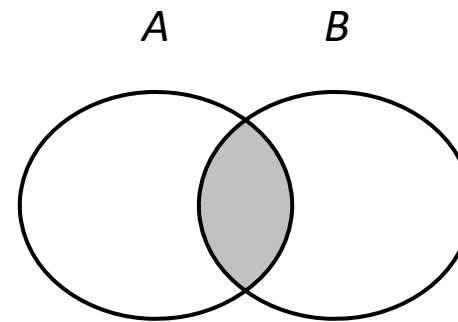
文氏圖

- ▶ The next two **Venn diagrams** depict the union and intersection of sets A and B .



(a) $A \cup B$

聯集



(b) $A \cap B$

交集

Sequences

序列：依某種順序將組成元素表列出來

- ▶ A **sequence** of objects is a list of these objects in some order.
 - ▶ We usually designate a sequence by writing the list within parentheses. 表列在小括號中
 - ▶ In a set the order doesn't matter, but in a sequence it does.
集合中：順序不重要 序列：順序是最重要的
- ▶ For example:
 - ▶ The sequence 7, 21, 57 would be written (7, 21, 57).
 - ▶ Hence (7,21,57) is not the same as (57, 7, 21).
(7, 21, 57) 與 (57, 7, 21) 不相等
- ▶ Repetition does matter in a sequence, but it doesn't matter in a set. 重覆在序列中是有意義的，但在集合中無意義
 - ▶ Thus (7,7,21,57) is different from (7,21,57) and (57, 7, 21).
不相同
 - ▶ The set {7, 21, 57} is identical to the set {7, 7, 21, 57}.
相同

Tuples and power sets

元組：有限長度的序列通常稱為“元組”

- ▶ Finite sequences often are called **tuples**.
 - ▶ A sequence with k elements is a k -tuple. k - “元組”
 - ▶ Thus (7,21,57) is a 3-tuple.
 - ▶ A 2-tuple is also called a **pair**. 2-“元組”又稱為“對”
- ▶ The **power set** of A is the set of all subsets of A .
- ▶ For example: 冪集合：A 的子集所形成的集合
 - ▶ If A is the set $\{0, 1\}$, the power set of A is the set $\{\emptyset, \{0\}, \{1\}, \{0, 1\}\}$.
 - $\emptyset$: 0個元素
 - $\{0\}$: 1個元素
 - $\{1\}$: 1個元素
 - $\{0, 1\}$: 2個元素
 - ▶ The set of all pairs whose elements are 0s and 1s is $\{(0, 0), (0, 1), (1, 0), (1, 1)\}$.
 - 0 和 1 所形成的 pair (,)
 - ↑ ↑
 - 可放 0 或 1

Cartesian product 迪卡兒積

- ▶ The **Cartesian product** of A and B , written $A \times B$, is the set of all pairs wherein the first element is a member of A and the second element is a member of B .

pair 所形成的集合 (放 A 的元素, 放 B 的元素) - 一般指 $A \times B$

- ▶ For example:

- ▶ If $A = \{1, 2\}$ and $B = \{x, y, z\}$, then

$$A \times B = \{(1, x), (1, y), (1, z), (2, x), (2, y), (2, z)\}.$$

- ▶ We can also take the Cartesian product of k sets, $A_1, A_2, \dots, A_k$, written $A_1 \times A_2 \times \dots \times A_k$. 2 個以上為特殊情形

- ▶ For example:

- ▶ If $A = \{1, 2\}$ and $B = \{x, y, z\}$, then

$$A \times B \times A = \{(1, x, 1), (1, x, 2), (1, y, 1), (1, y, 2), (1, z, 1), (1, z, 2), \\ (2, x, 1), (2, x, 2), (2, y, 1), (2, y, 2), (2, z, 1), (2, z, 2)\}.$$

Outline

- ▶ Mathematical Notions and Terminology
- ▶ **Definitions, Theorems, and Proofs**
- ▶ Types of Proof

Definitions, theorems, and proofs

- ▶ Theorems and proofs are the heart and soul of mathematics, and definitions are its spirits.

定理和證明是數學的靈魂,定義是數學的精神

- ▶ These three entities are central to every mathematical subjects, including algorithms.

對於任何數學科目都很重要,包括演算法

- ▶ In this class, you are expected to pick up the ability of reading and writing a concrete proof.

目標: (1) 讀證明

(2) 寫證明

Definitions 定義是用來描述事或物, 以及要使用的符號

- ▶ **Definitions** describe the objects and notions that we use.
 - ▶ Precision is essential to any mathematical definition. 精確是最重要的
 - ▶ When defining some object we must make clear what constitutes that object and what does not.
描述事或物時, 要清楚說明包含與不包含的東西
- ▶ For example:
 - ▶ A **set** is a group of objects represented as a unit.
 - ▶ The objects in a set are called its **elements** or **members**.
 - ▶ A **tree** graph is a connected graph without cycles.

Proofs 在定義事或物以及符號後, 我們會以“數學敘述”來描述一些性質

- ▶ After we have defined various objects and notions, we usually make **mathematical statements** about them.
 - ▶ A statement expresses that some object has a certain property.
 - ▶ The statement may or may not be true. 可能錯也可能正確
 - ▶ The statements must be precise without any ambiguity.
一定要精確, 不可模稜兩可
- ▶ A **proof** is a convincing logical argument that a statement is true. 有說服力的邏輯論證, 來說明 statement 是對的
 - ▶ In mathematics an argument must be airtight, that is, convincing in an absolute sense. 論證是無懈可擊的
 - ▶ A mathematician demands proof beyond any doubt.
禁得起任何質疑

Theorems, lemmas, and corollaries

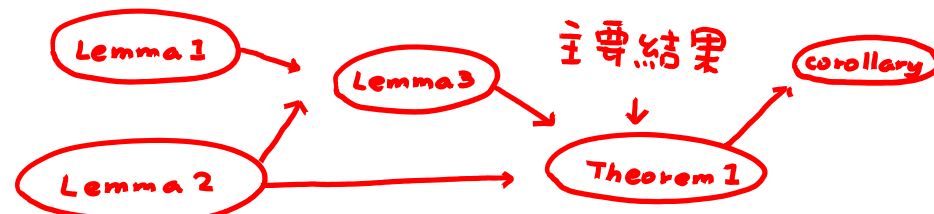
- ▶ A **theorem** is a mathematical statement proved true. Generally we reserve the word for statements of special interest. 定理：是證明真實的一個數學敘述

- ▶ **Lemmas** are the proved statements that are interesting only for their assistance in the proof of another statement.

引理：用來輔助定理

- ▶ Occasionally a theorem or its proof may allow us to conclude easily that other, related statements are true. These statements are called **corollaries** of the theorem.

corollary 推論：由主要性質容易推導的性質
(結果)



Finding Proofs 證明是唯一 - 可以決定 statement 是否正確的方法

- ▶ The only way to determine the truth or falsity of a mathematical statement is with a mathematical proof.
- ▶ Unfortunately, finding proofs isn't always easy.
 - ▶ It can't be reduced to a simple set of rules or processes.
- ▶ However, some helpful general strategies are available.
 - ▶ Carefully read the statement you want to prove. 敘述要了解
 - ▶ Make sure you understand all the notation. 符號要懂得
 - ▶ Rewrite the statement in your own words. 以自己的話重寫敘述
 - ▶ Break it down and consider each part separately. 將步驟分步去思考
 - ▶ Experimenting with examples. 舉出例子
 - ▶ Try to find an object that fails to have the property, called a **counterexample**. 去尋找反例, 若無反例則可確定為是

Multipart statements_{1/2}

多方敘述 Ex: $A \Leftrightarrow B$

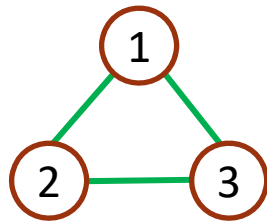
- ▶ One frequently occurring type of **multipart statement** has the form " P if and only if Q ", often written " P iff Q ".
 - ▶ Both P and Q are mathematical statements.
 - ▶ The first part is " P only if Q ," which means: If P is true, then Q is true, written $P \Rightarrow Q$. 若 P 則 Q , 若 P 為真則 Q 成立
 - ▶ The second is " P if Q ," which means: If Q is true, then P is true, written $P \Leftarrow Q$. 若 Q 則 P , 若 Q 為真則 P 成立
 - ▶ We write " P if and only if Q " as $P \Leftrightarrow Q$.
若且唯若
 - ▶ To prove a statement of this form you must prove each of the two directions.

Multipart statements_{2/2}

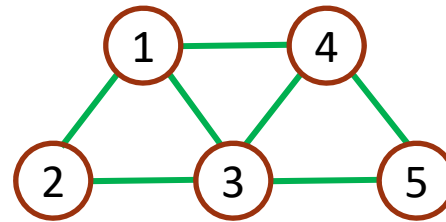
- ▶ Another type of multipart statement states that two sets A and B are equal. 若 A, B 集合相等
 - ▶ The first part states that A is a subset of B . 先證 A 為 B 的子集 ($A \subseteq B$)
 - ▶ The second part states that B is a subset of A . 再證 B 為 A 的子集 ($B \subseteq A$)
- ▶ To show that $A = B$, we must prove the following two statements.
 - ▶ Every member of A also is a member of B .
 - ▶ Every member of B also is a member of A .
每一個在 A 的 element 也一定在 B 裡 ($A \subseteq B$)
每一個在 B 的 element 也一定在 A 裡 ($B \subseteq A$)

An example for finding Proofs_{1/2}

- ▶ Suppose that you want to prove the statement
“For every graph G , the sum of the degrees of all the nodes in G is an even number”.
證明 degree 和為偶數 (degree 為與此點相鄰的點個數)
- ▶ **First**, pick a few graphs and observe this statement.



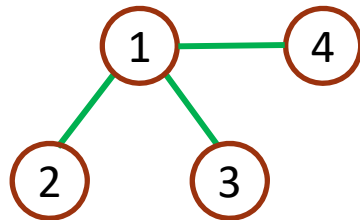
$$\begin{aligned}\text{sum} &= 2 + 2 + 2 \\ &= 6\end{aligned}$$



$$\begin{aligned}\text{sum} &= 3 + 2 + 4 + 3 + 2 \\ &= 14\end{aligned}$$

An example for finding Proofs_{2/2}

- ▶ **Next**, try to find a counterexample, that is, a graph in which the sum is an odd number. 尋找反例



Every time an edge is added,
the sum increases by 2.

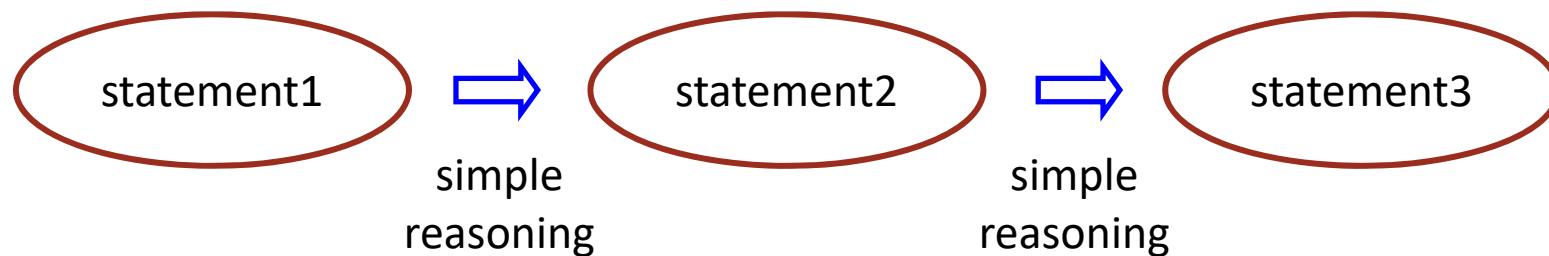
每增加一條邊，degree總和多2

- ▶ Can you now begin to see why the statement is true and how to prove it?
- ▶ If you are still stuck trying to prove the statement, try to prove a **special case** of the statement. 若仍無法證明，則找尋特例
- ▶ First try for $k = 1$, as well as $k = 2$, $k = 3$, $k = 4$ and so on.

Writing proofs

- ▶ A well-written proof is a sequence of statements, wherein each one follows by simple reasoning from previous statements in the sequence.

一個好的證明：為一連串的敘述，且下一個敘述可由上一個敘述簡單接得



- ▶ Carefully writing a proof is important. 寫出證明的好處
 - ▶ Enable a reader to understand it. 使讀者易於理解
 - ▶ Make sure that it is free from errors. 避免出錯

Tips for producing a proof

- ▶ Be patient. 有耐心
 - ▶ Researchers sometimes work for weeks or even years to find a single proof.
- ▶ Come back to it.
 - ▶ Look over the statement you want to prove, think about it a bit, leave it, and then return a few minutes or hours later.
- ▶ Be concise. 小心且精確, 清楚簡要的表達出來
 - ▶ Good mathematical notation is useful for expressing ideas concisely.
 - ▶ But be sure to include enough of your reasoning when writing up a proof so that the reader can easily understand what you are trying to say.

證明 degree 和為偶數 (degree 為與此點相鄰的點個數)

An example for producing a proof

► Theorem : **For every graph G , the sum of the degrees of all the nodes in G is an even number.**

► Proof.

- (I) ► Every edge in G is connected to two nodes. 每個 edge 需連接 2 個點
- (II) ► Each edge contributes 1 to the degree of each node to which it is connected. 連接到這條邊的點, degree 增加 1
- (III) ► Therefore each edge contributes 2 to the sum of the degrees of all the nodes. 一個 edge 會使 degree 增加 2 (由 I. II 得知)
- (IV) ► Hence, if G contains e edges, then the sum of the degrees of all the nodes of G is $2e$, which is an even number. 若有 e 條 edge
⇒ degree 和為 $2e$

Outline

- ▶ Mathematical Notions and Terminology
- ▶ Definitions, Theorems, and Proofs
- ▶ **Types of Proof**

Types of proof

- ▶ Several types of arguments arise frequently in mathematical proofs.
 - ▶ Proof by construction. 建構法
 - ▶ Proof by contradiction. 反證法
 - ▶ Proof by induction. 歸納法
- ▶ Note that a proof may contain more than one type of argument. 證明中可能會有好幾種方法,因為會包含許多子證明
 - ▶ Because the proof may contain within it several different subproofs.

Proof by construction_{1/2}

- ▶ A graph is said to be **k -regular** if every node in the graph has degree k . 每個 node 的 degree 都是 k

- ▶ Theorem : For each even number n greater than 2, there exists a 3-regular graph $G(V, E)$ with n nodes.

- ▶ 大於 2 個 node 的所有偶數個 node, 必有一種畫法使得每個的 degree 為 3
- ▶ Proof.

- ▶ Let n be an even number greater than 2.

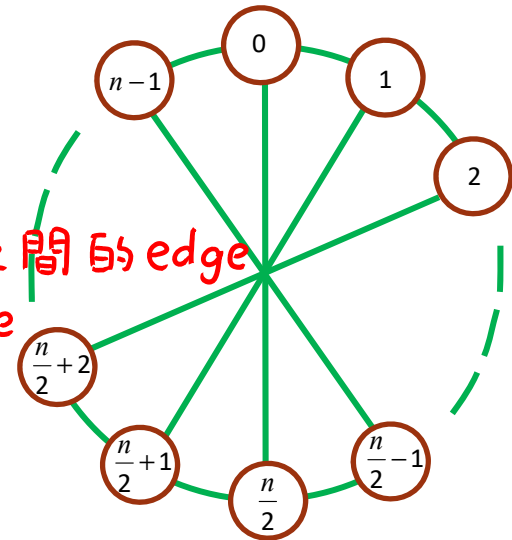
- ▶ The set $V = \{0, 1, \dots, n-1\}$. $0 \sim (n-1)$ 個點

- ▶ The set $E = \{(i, i+1) \mid \text{for } 0 \leq i \leq n-2\}$ $0 \sim (n-1)$ 之間的 edge

$$\cup \{(n-1, 0)\} \quad (n-1) \sim 0 \text{ 之間的 edge}$$

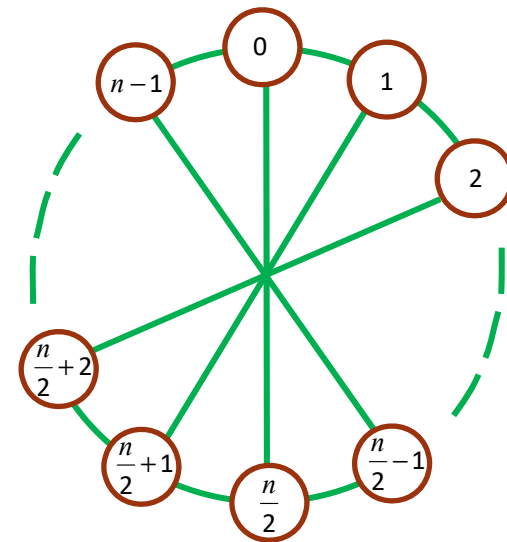
$$\cup \{(i, i+n/2) \mid \text{for } 0 \leq i \leq n/2-1\}.$$

i 與 $(i+n/2)$ 之間的 edge



Proof by construction_{2/2}

- ▶ The set $E = \{(i, i+1) \mid \text{for } 0 \leq i \leq n-2\}$
 $\cup \{(n-1, 0)\}$
 $\cup \{(i, i+n/2) \mid \text{for } 0 \leq i \leq n/2-1\}$.
- ▶ The edges described in the top and middle lines of E go between adjacent pairs around the circle.
- ▶ The edges described in the bottom line of E go between nodes on opposite sides of the circle.
- ▶ This mental picture clearly shows that every node in G has degree 3.



Proof by contradiction 反證法

- (I) ▶ In this kind of arguments, we first **assume** that the theorem to prove is **false**. 首先假設是錯的
- (II) ▶ Then we show that this assumption leads to an obviously false consequence, called a **contradiction**. 最後發現一個矛盾的結果 (與事實相反)
 - ▶ We use this type of reasoning frequently in everyday life.
 - ▶ Jack just came in from outdoors and he is completely dry.
 - ▶ We want to prove that It's not raining.
 - ▶ If it were raining, Jack would be wet.

(assume false)(contradiction)
 - ▶ Therefore, it must not be raining.

An example for proof by contradiction

- ▶ A number is **rational** if it is a fraction m/n where m and n are integers. 有理數可表示成 $\frac{m}{n}$
- ▶ Theorem: $\sqrt{2}$ is **irrational**. 證明 $\sqrt{2}$ 為無理數
- ▶ Proof.
 - ▶ Assume for a contradiction that $\sqrt{2} = \frac{m}{n}$ is rational, where m and n are relatively prime integers.
 - ▶ $\sqrt{2} = \frac{m}{n} \Rightarrow n\sqrt{2} = m \Rightarrow 2n^2 = m^2$.
互質
 - ▶ Since m^2 is even, m must be even as well. m^2 為偶數 $\Rightarrow m$ 必為偶數
 - ▶ Let $m = 2k$. Then we have $2n^2 = (2k)^2 = 4k^2$, which implies $n^2 = 2k^2$.
 - ▶ Since n^2 is even, n is even.
 - ▶ Both m and n are even, a **contradiction** to our assumption.

與假設矛盾

Proof by induction 歸納法

- ▶ An advanced method to show that **all elements of an infinite set** have a specified property.
證明無限集中的元素都有某種特殊性質的方法
 - ▶ Suppose that our goal is to prove that $P(k)$ is true for each natural number $k \in \{1, 2, 3, \dots\}$.
 - ▶ The format for writing down a proof by induction is as follows.
 - ▶ **Basis:** Prove that $P(1)$ is true.
 - ▶ **Induction step:** For each $i \geq 1$, assume that $P(i)$ is true and use this assumption to show that $P(i+1)$ is true. 假設 $P(i)$ 為是 $\Rightarrow P(i+1)$ 也為是 ($i \geq 1$)
 - ▶ $P(1)$ is true: basis.
 - ▶ $P(2)$ is true: $P(1)$ is true + induction step.
 - ▶ $P(3)$ is true: $P(2)$ is true + induction step.
 - ... $P(k)$ is true
-

An example for proof by induction

- ▶ For each $n \geq 1$, we have

$$1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}.$$

- ▶ Proof.

- ▶ **The basis:** For $n = 1$, $1^2 = 1 = \frac{1(1+1)(2 \times 1 + 1)}{6}$.

- ▶ **Induction step:** For each $k \geq 1$, assume that the formula is true for $n = k$ and show that it is true for $n = k + 1$.

$$\begin{aligned} \sum_{i=1}^{k+1} i^2 &= \sum_{i=1}^k i^2 + (k+1)^2 = \frac{k(k+1)(2k+1)}{6} + (k+1)^2 = \frac{k(k+1)(2k+1) + 6(k+1)^2}{6} \\ &= \frac{(k+1)(2k^2 + 7k + 6)}{6} = \frac{(k+1)(k+2)(2(k+1)+1)}{6} \end{aligned}$$